MODEL QUESTION PAPER

Revision 2010
TED (10) 4097

Fifth Semester Diploma Examination In Polymer Technology

PLASTIC TECHNOLOGY

Maximum Marks ;100 Time 3Hours

PART –A(10marks)

1 . Answer the following questions in one or two sentences. Each question carries 2 marks.

a). Distinguish plasticizers and lubricants with examples?
b). What are Eco-friendly plastics ? Give two examples.
c)Why Z-N catalyst is used for the production of HDPE?
d)Name the monomers of PET & Nylon 6,6 ?
e)Pressure cooker handles are usually produced from PF . Why? (5x2=10)

PART-B
(30 marks)

11. Answer any five full questions from the following Each question carries 6 marks .

a). (1) Differentiate Tg &Tm of plastic materials .Give its importance (3)
(2) Why coupling agents & fillers are incorporated in a plastic compound (3)
b. (1) What are anti degradents ? Give two examples .? (3)
(2) What is compounding of plastics ? Name two compounding equipments (3)
c. (1) How the monomer vinyl chloride is prepared ? (3)
(2) What is tacticity of poly propylene? Write structures of 3 forms of PP. (3)
d. (1)What is HIPS ? How it is prepared / (3)
(2) How the monomer MMAis prepared (3)
e HMDA& adipic acid are the monomers of nylon6,6 how they are prepared (6)
f. (1) Differentiate thermosetting polyester and thermoplastic polyester (3)
2)Suggest suitable plastics for the following products (3)
i)Transparent sheets ii)Gears iii) Compact Discs
g (1) What are the monomers of Epoxy resin ? write their structures (3)
(2) What are A stage, B stage and C stage resins in thermo setting polymers (3)
(5x6=10)

PART-C
(60marks)

Answer one full question from each module. Each question carries 15 marks

MODULE -I

III (a) What are the advantages and disadvantages of plastics over other conventional materials like metals and ceramics (8)
(b) Properties of a product can be altered by compounding. Justify with a suitable example (7)

OR

IV(a) Classify the compounding ingredients of plastic products . Give its function and examples of each ingredients ? (8)
(b) Classify plastics according to their temperature response ,origin &chemical nature with examples (7)

MODULE-II

V (a) Explain the manufacture of LDPE and Compare its properties with HDPE (8)

(b) Explain the manufacture of Poly propylene ,starting from its monomer preparation Mention its properties and application (7)

OR

VI (a)Explain the industrial production of PVC with typical formulations and block diagram ? Mention its properties and application (8)

(b)Explain the Tower Process for manufacture of Poly Styrene with neat diagram Mention its properties and application (7)

MODULE-III

VII (a) Explain the raw materials, Polymerization Properties and application of Poly acetal resin (8)

(b)Distinguish the production properties and application of Nylon 6. &Nylon6,6 (7)

OR

VIII (a) Explain the raw materials, Polymerization Properties and application of PET (8)

(b) Describe the raw materials, Polymerization Properties and application of Polycarbonate (7)

MODULE-IV

IX (a) Explain the manufacture of Epoxy resin ,starting from its monomer preparation Mention its properties and application (8)

(b) Explain the manufacture molding powder which is used for electrical Application. (7)

OR

X (a) Explain the manufacture of Novalac PF resin with typical formulation and neat diagram (8)

(b) Describe the raw materials, Polymerization Properties and application of Silicone plastics (7)

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